

FALCON's Proposal to Unique Health Products for Supply of Neo Automation System.

November 7, 2025

Doc. No. - Sales/FR/PT/007/R0



Kind Attention –

Mr. Cameron Walkerden

Unique Health Products

Offer Ref: F24-00543_R2; Date: 07-11-2025

Subject – Techno -Commercial Offer for Supply of NEO Automation System.

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have done an in-depth data analysis and evaluated various solution options best suited for your requirements, along with the information collected during the meetings and discussions with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to being your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Intra-logistics Automation Technologies and references.

To conclude, I would like to add my personal commitment on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

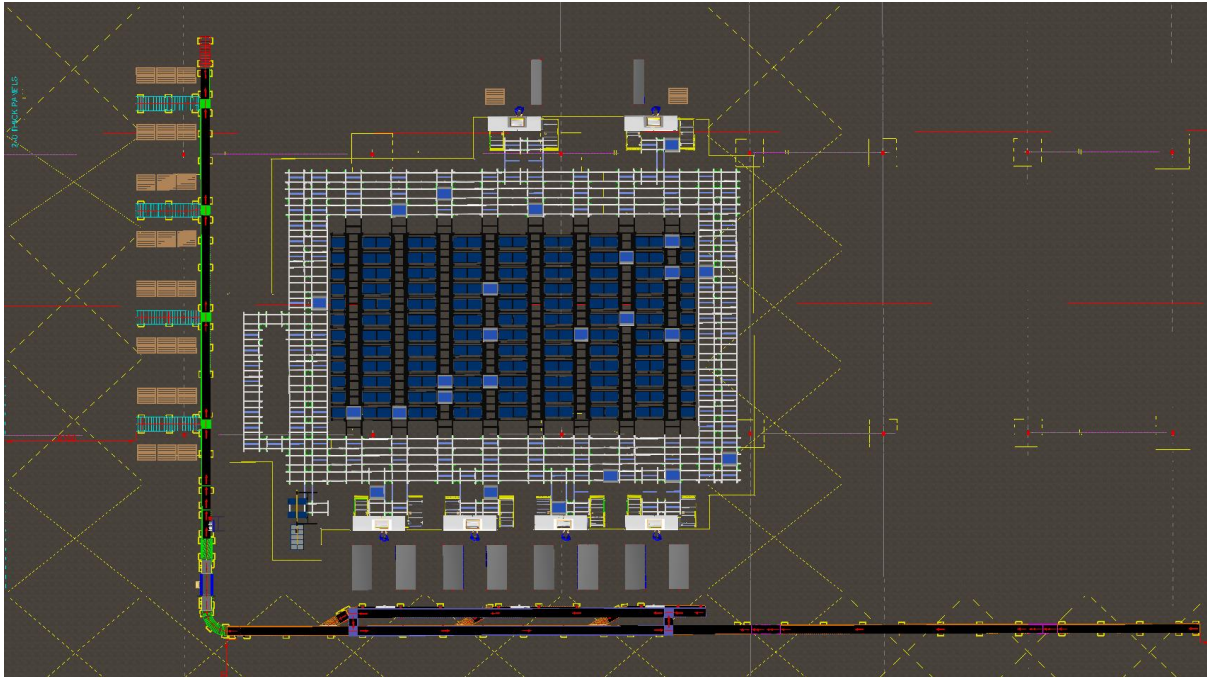
Best Regards,

Chris Josey

Regional Head ANZ

Response to Unique Health Products for Supply of Neo Automation System

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Proposal Reference – F24-00543_R2

Falcon Autotech Private Limited

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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quotation
2	PPH	Shipments Per Hour
3	ICR	Intelligent Character Recognition
4	MEZZ	Mezzanine
5	FWD	Friction Wheel Drive
6	ECDS	Empty Carrier Detection System
7	AC	Alternating Current
8	DC	Direct Current
9	PLC	Programmable Logic Controller
10	IT	Information Technology
11	BOQ	Bill Of Quantity
12	I/O	Input/ Output
13	PDP	Power Distribution Panel
14	PC	Personal Computer
15	UPS	Uninterrupted Power Supply
16	CBS	Cross Belt Sorter
17	MDR	Motor Driven Roller
18	VDS	Volume Distribution System
19	IPP	Individual Productivity Potential
20	VM	Virtual Machine
21	ANZ	Australia & New Zealand
22	FOC	Free of Cost
23	CBS	Cross Belt Sorter
24	CEP	Courier Express Parcel
25	DAP	Design Approval Phase
26	UHP	Unique Health Products

2. Executive Summary

Falcon Autotech is pleased to confirm its great interest in responding to this RFQ of NEO Automation System for your QLD location. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening, understanding your needs, and ensuring this project's success. Following the same objective for this NEO system, we are happy to offer a compliant solution meeting all technical and operational requirements at competitive price, delivering key results.

Our solution is based on the following key characteristics:

NEO automation with automated order takeaway, packing and sortation system: -

- *The NEO System is designed for a capacity to accommodate 4224 totes. This is being calculated as per the 3-week average data shared with us. Pro rating this data we have calculated for 4 weeks average as well, thus NEO with 5544 tote locations is also quoted as an option.*
- *4 Pick Stations and 2 Put stations are provided for simultaneous operations. The put stations can also be used as pick stations if needed. One dedicated conveyor based put station is provided for fragile items put away.*
- *24 NEO bots for picking and putting away functions.*
- *Integrated battery charging station for NEO bots.*
- *Put to light order setup for order consolidation and put-away consolidation.*
- *Separate infeed lines with integrated auto carton sealing machines in front of the order consolidation area. Empty order carton accumulation conveyors are provided above this system.*
- *Highway conveyor line for order cartons integrated with side barcode scanning, weighing and auto label printing system.*
- *Dedicated conveyors for cartons from bulk area merging into the highway conveyor.*
- *Route wise sortation with 4 output destinations.*
- *Loop Conveyor for empty cartons.*

This solution has been crafted especially keeping UHP's technical and operational requirements, as discussed during various meetings, making it a tailor-made solution delivering a faster TAT and an efficient material flow.

*The Falcon Autotech automations systems are proudly **MADE IN INDIA**. They incorporate in-house designed and developed system components and associated peripherals.*

1. Falcon's reliable automation systems

Falcon Autotech has over 100 automation system installed globally ranging from small and large sortation systems to mid and large sized ASRS systems, used by most innovative brands such as Aramex, Couriers Please, TGE, Decathlon, Zepto, Amazon, Flipkart Delhivery, Asendia, Fastway and many more. The main and critical components of the Falcon Autotech systems, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from industry leading suppliers in the world, such as SEW, Siemens, SICK, Vahle, Forbo etc. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. Commitment to quality systems

Demonstrating Falcon's clear commitment to the UHP's satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live.

Our expertise in executing world-class, projects across the ANZ region has been thoroughly demonstrated and validated. We offer:

- A proven history of success and broad experience in implementing and maintaining automated systems.
- ISO 9001 Quality Assurance certification, ensuring consistent quality across our products, processes, and outcomes.
- Falcon Warehouse Control System (WCS) is designed to meet the real-time operational needs of automated materials handling systems. WCS can be molded based on the client's requirements to enhance operations.
- Comprehensive service and warranty programs. We deliver both operational and emergency support with unmatched responsiveness and availability throughout the region.
- Strong financial stability, backed by a long-standing track record.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 12 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0



Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces



Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS



Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5 Product Lines

15+ Countries with Live Installations

1800+ Total Installed Systems Globally

1Million+ Sorts Per Hour

600+ Employees

Long-standing Relationships with industry leaders

Ecom Retail

Flipkart

amazon

Lazada

مركز

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

G

SWIGGY

TITAN

Mind Garden Limited

News & Articles



FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.



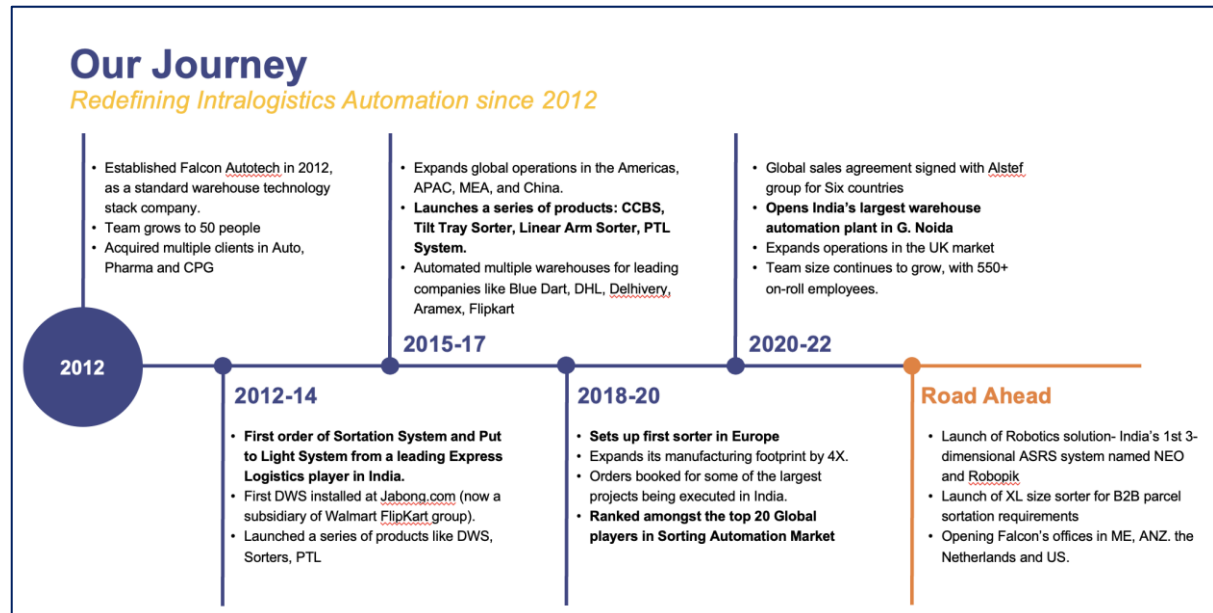
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company; our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. To be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and delivering world-class projects in return.



Product and Solutions

With 100% focused on Parcels, Eatches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Popup Sorter
- Sweep Sorter
- Pusher Sorter



PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
 - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
 - Mini (600MM)
 - Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by





Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our In house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

























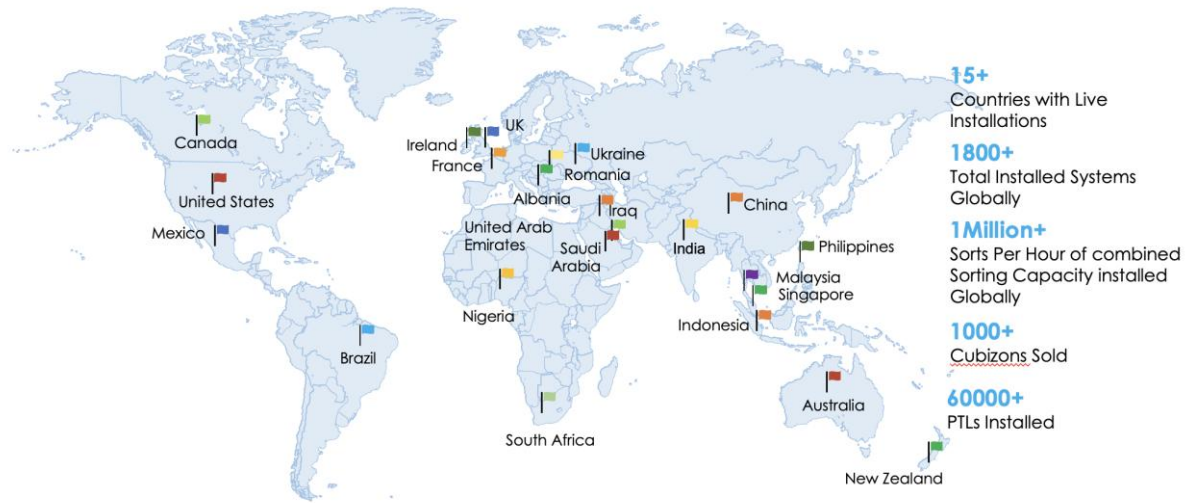






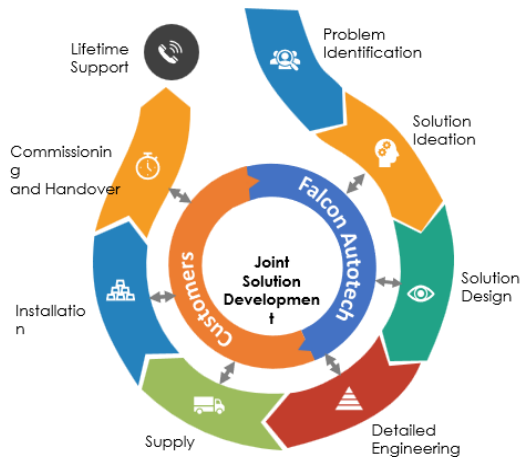
With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

Global Market Presence



Customer Engagement Model

Engaging and supporting the customer throughout the solution lifecycle



Helps Build Trust with Customer

Collaborative Process

Full Transparency

Increased Engagement

Less Solution Defects

4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



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FALCON AUTOTECH

Falcon Autotech expands its wings by opening its office in The Netherlands, Europe

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NEW DELHI, India and AMSTERDAM, April 15, 2024 (GLOBE NEWSWIRE) -- Falcon Autotech (Falcon), a leading global intralogistics automation solutions provider, opens its



Breaking Barriers and Optimizing Efficiency – Journey of Delivering India's Largest Sortation System

"It was a bold decision. If it didn't work, our company would have shut down," says Naman Jain, founder and CEO of warehouse automation startup Falcon Autotech, referring to a big gamble he had taken more than a decade ago. The year was 2013, Jain and his team were negotiating a deal with a major e-commerce logistics company that had just placed an updated order for a sorter with a capacity of 6,000 parcels per hour. That was 4x the specification earlier agreed upon.

"All the founders came together, and we took a calculated risk," recalls Jain. At the time, Falcon did not have the technology to develop such sorters, and hence, the said client refused to pay any money upfront and offered to pay the machine's price in monthly installments. However, there was a caveat: if it failed even once, Falcon would simply take the machine back and refund the entire amount. Falcon's 'calculated risk' paid off. The machine worked flawlessly and was handed over to the parcel company after 36 months. "We are now doing projects worth INR100 crore. That was unthinkable 10 years back," says Jain.

Falcon Autotech has covered a lot of ground since that big bet. Last month, it installed India's largest sortation equipment at



Economic Times Features Falcon Autotech

Naman Jain
Chief Executive Officer

From sorters to conveyors & robot based systems, here's the top tech warehouses are investing in

India's warehousing industry has travelled a long distance from "godowns" to modern storage facilities called Grade A warehouses.

Warehouse automation is a gradual process. India started off late, but that may actually help it bypass some of the mid-age technologies and adopt the latest ones. While several players are still warming up to automation, which are the



Falcon Announces Successful Go-Live of Automated Parcel Sortation Solution at DTDC's Chennai Facility

New Delhi, India – 2nd Aug 2023: Falcon Autotech, a leading supplier of intralogistics automation solutions, has been selected by DTDC Express Ltd, one of India's leading integrated express logistics company, to automate its parcel sorting operations at its super hub of 1,75,000 sq ft in Chennai, Tamil Nadu. Using its cross belt sorter technology, Falcon has designed DTDC's parcel sorting system, which can handle 6,000 parcels per hour, operate in a 24 X 7 environment, and can be expanded to cater to future growth.

The new linear cross-belt solution leverages cutting-edge technology to automate key sorting processes in DTDC's warehouse, including parcel profiling and sorting. This solution is designed to optimize the space requirements for sorting operations, increase efficiency, and reduce operational costs.

"We are thrilled to see our warehouse automation solution go live with DTDC, this solution is a testament to our commitment to providing innovative intra-logistics solutions that meet the evolving needs of our customers," said Falcon's CEO Naman Jain.



Transforming Indian Retail: The Impact of ASRS on Warehousing Operations

In recent years, the Indian retail industry has experienced unprecedented growth, propelling the nation to become the fifth-largest global retail destination. The industry, marked by its resilience and adaptability, went through significant transformations during the pandemic. While online shopping saw a remarkable surge, the reopening of physical stores ushered in a resurgence of the multi-sensory shopping experience. As a testament to this growth, shopping malls now encompass an astonishing 23.25 million square feet of retail space. However, the evolving consumer preferences have placed substantial pressure on businesses, necessitating the seamless integration of e-commerce and in-store experiences, which, in turn, has led to complex logistics challenges. The traditional warehousing systems struggled to cope with the demand for faster processing and the requirement for expanded storage capacities.

Historically, retailers had heavily relied on manual labor for their warehousing operations. However, the dynamism of today's retail market demands a shift towards automation. Enter Automated Storage and Retrieval Systems (ASRS), a technological

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

5.1 Project 1- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.2 Project 2- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.3 Project 3- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- 2 Sets of Double Decker CBS Sorters
- Mezzanine Structures
- Automated Singulators
- Fully Automatic Inductions
- Semi-Automatic Inductions
- Telescopic belt conveyors
- PVC Belt Conveyors
- Modular Belt Conveyors
- Spiral Chutes with Braking rollers
- 5-Sided Scanning Tunnels
- High speed weighing conveyors
- Direct Bagging Chutes
- Put to Light Chutes
- Volume Distribution systems
- High Availability Server Systems
- WCS

Site Pictures –



5.4 Project 4- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

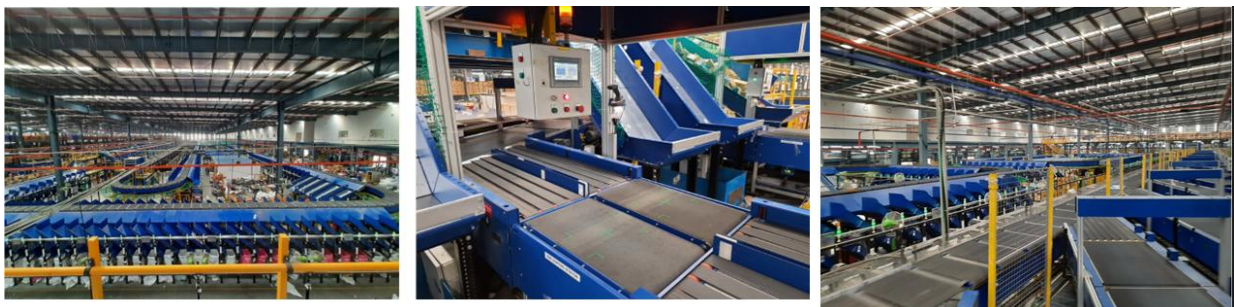
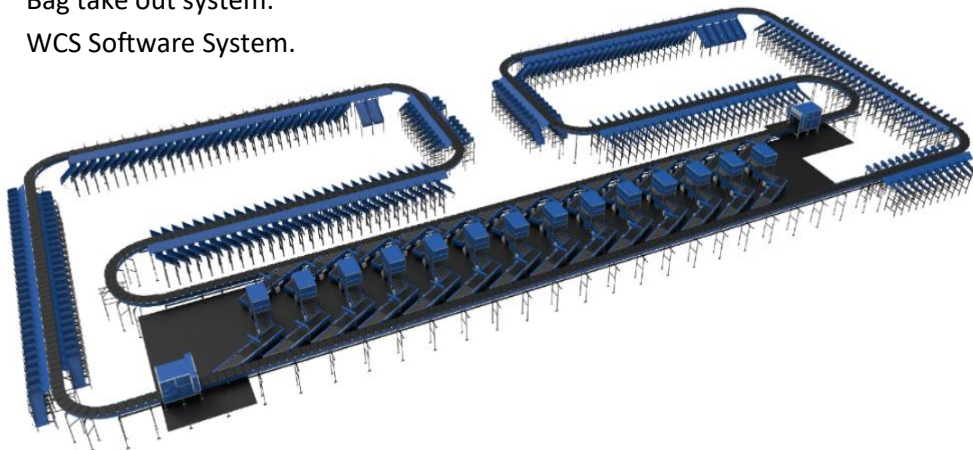
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system.
- WCS Software System.



5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

The customer chose Falcon Autotech based on its unique design, capability of seamless integration with WMS and life cycle support services.

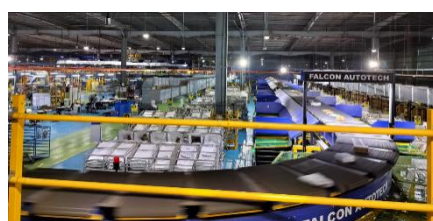
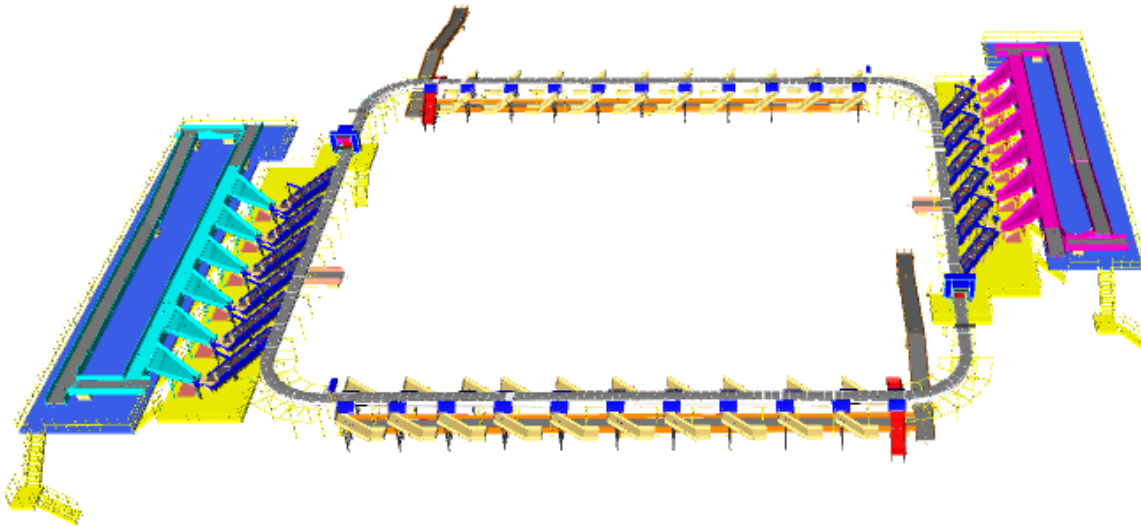
Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 40 Collection Type Chutes

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Collection type chutes
- Bag take out system.
- WCS Software System.

Layout and Site Pictures -



5.6 Project 6- (Client – E-Commerce, India)

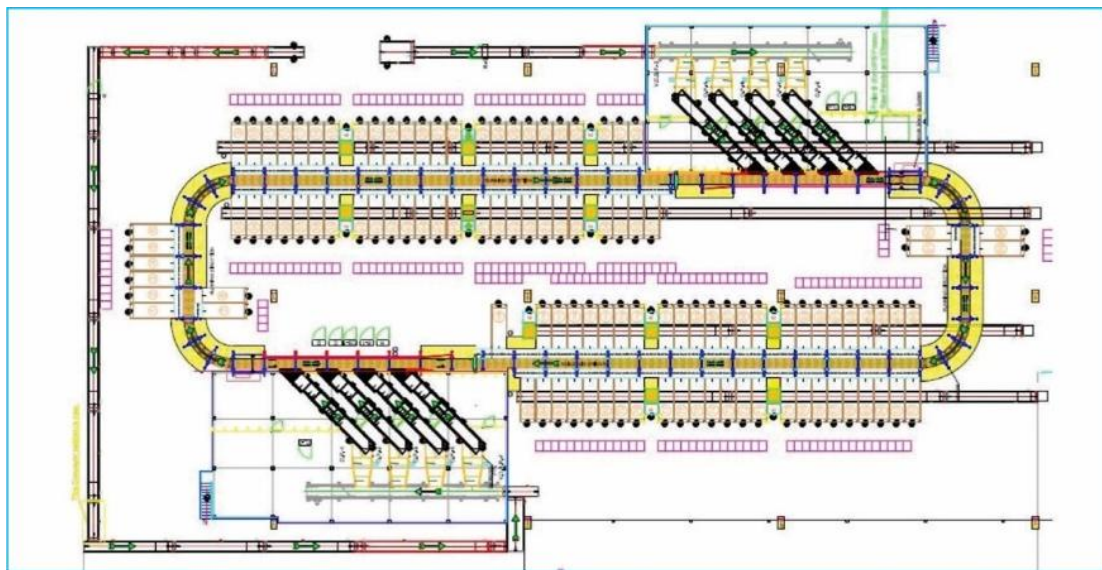
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



5.7 Project 7- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of shipments arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Shipment handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5.8 Project 8- (Client – E-Commerce, UAE & Saudi Arabia)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 shipments per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –



6. Handled Shipment Spectrum

The storage capacity for the NEO is calculated based on the stock summary shared with us, SKUs has been analysed for their dimensions and are considered for NEO only if they are smaller than the size of the NEO tote i.e. 760 X 520 X 410 MM and at the same time can fit in 4 or more quantity in one tote.

The allowed range of cartons that can be stored and conveyed on the conveyor and sortation systems is mentioned below:

Boxes/Cartons/Totes		
Specification	Unit	Value
Max Length	mm	470
Max Width	mm	315
Max Height	mm	285
Max Weight	Kg	20
Min length	mm	270
Min Width	mm	220
Min Height	mm	160
Min Weight	gm	200

**Please note that the box sizes are considered as per the box sizes shared by UHP. Any changes in these box sizes may lead to design changes.*

6.1 Shipments to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Items must not have magnetic content, otherwise the shipment behavior cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non-conveyable items.
4. Shipments should be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system, they must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the conveyor belt).
9. The shipments must have at least one flat and a regular surface providing enough stability.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.2 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non-conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, like spherical/cylindrical item.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points or edges (e.g., Nails), that can damage the conveyors or the sorter.
5. Fragile shipments with content not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Inadequately packed items that could be damaged during automatic transportation.
11. Electrostatically loaded items.
12. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non-conveyable items.

6.3 Barcode specifications

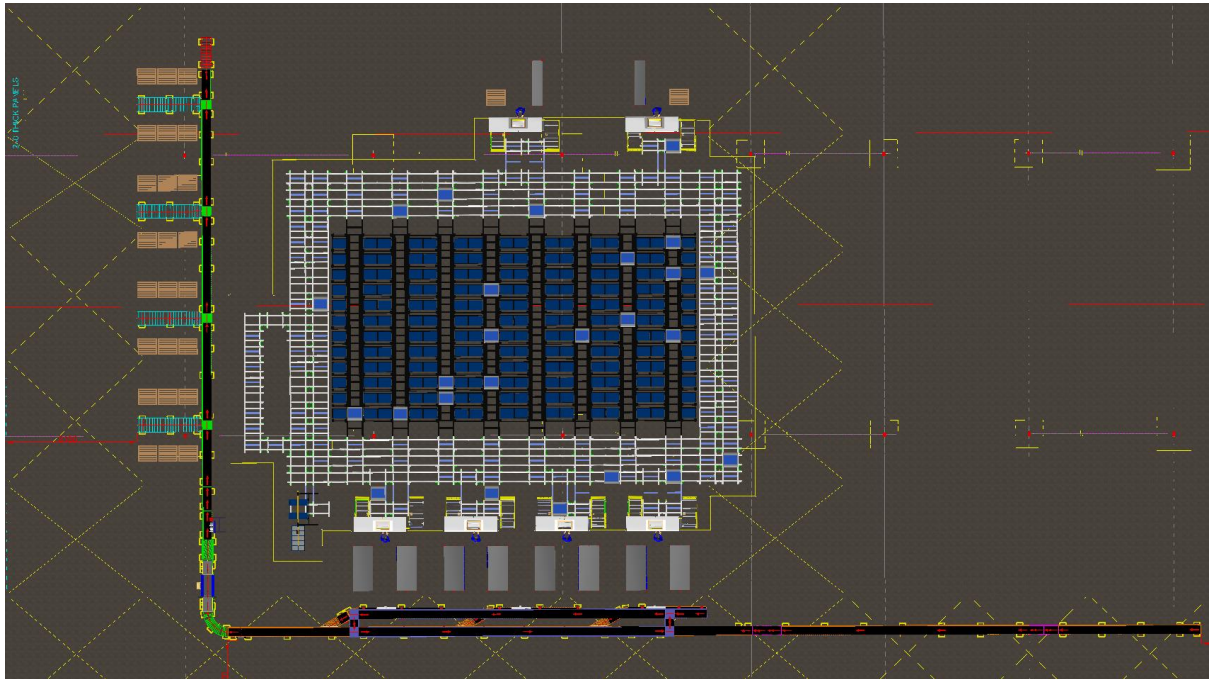
1. Min. Code Resolution for 1D codes to be 0.4 mm (15mil) or higher and Min. 1D Code Height to be 15 mm.
2. Quiet zone of 1D barcode to be at least ten times to the width of smallest bar.
3. Min. Code Resolution for 2D codes to be 0.80 mm or higher and 2D code size of 30 X 30 mm or higher.
4. Bar code quality must be in accordance with ANSI Grade A or B standards for the camera solution.
5. Codes are visible for the ident system (not crumpled, shadowed, no direct reflection or hidden by objects, etc.)

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the NEO automation system, as per UHP's requirements.

7.2 Summary of the System



Major components of the proposed solution-

1. **Neo Grid:** The Neo is designed for 4244/5544 tote locations. Second option with 5544 tote locations is also provided.
2. **Pick Stations:** 4 Pick Stations are provided on one edge of the system. Each pick station is associated with 24 order locations catered with a Put to light secondary sorting system.
3. **Put Stations:** 2 Put Stations are provided on the other edge of the system. These stations can be used for picking also in case not put away is happening at that time. 12 Put to light locations are provided at each station for consolidation of inbound material.
4. **Auto Carton Sealers:** 3 Auto carton sealer machines are provided with separate infeed conveyors to cater to the desired throughput.
5. **Empty Order Carton Accumulation Loop Conveyor:** A loop accumulation conveyor is provided at 1600 MM height for empty cartons staging.
6. **Auto Scanning, Weighing and Labelling System:** Completed order cartons are scanned, weighed and label pasted automatically.
7. **Conveyor for Bulk Area:** Dedicated Conveyor for cartons in bulk area merging into the main highway conveyor.
8. **Route Wise Sortation System:** An automated route wise sortation system is provided with 4 output locations. A dedicated output is provided for rejects and sort fails.

7.3 Neo ASRS System:

Particular	Value
Total Storage (Totes)	4244/5544
Number of Rack Levels	22
Number of Orders Per Day	576
Number of Units Per Day	45,905
Average Number of Eaches Per SKU	13.65
Number of Hours Per Day - Productive	6-8
Number of GTP Pick Stations Considered	4
Number of Orders Per Hour	96
Number of Bots Required for Picking	16
Max Units for Put Away	45,905
Assumed Average Units Per Bin	50
Number of Put Stations Considered	2
Number of Bots Required for Put Away	8
Total GTP Stations Considered	6
Total Bots Required	24
Number of PTL/Pick Station	24

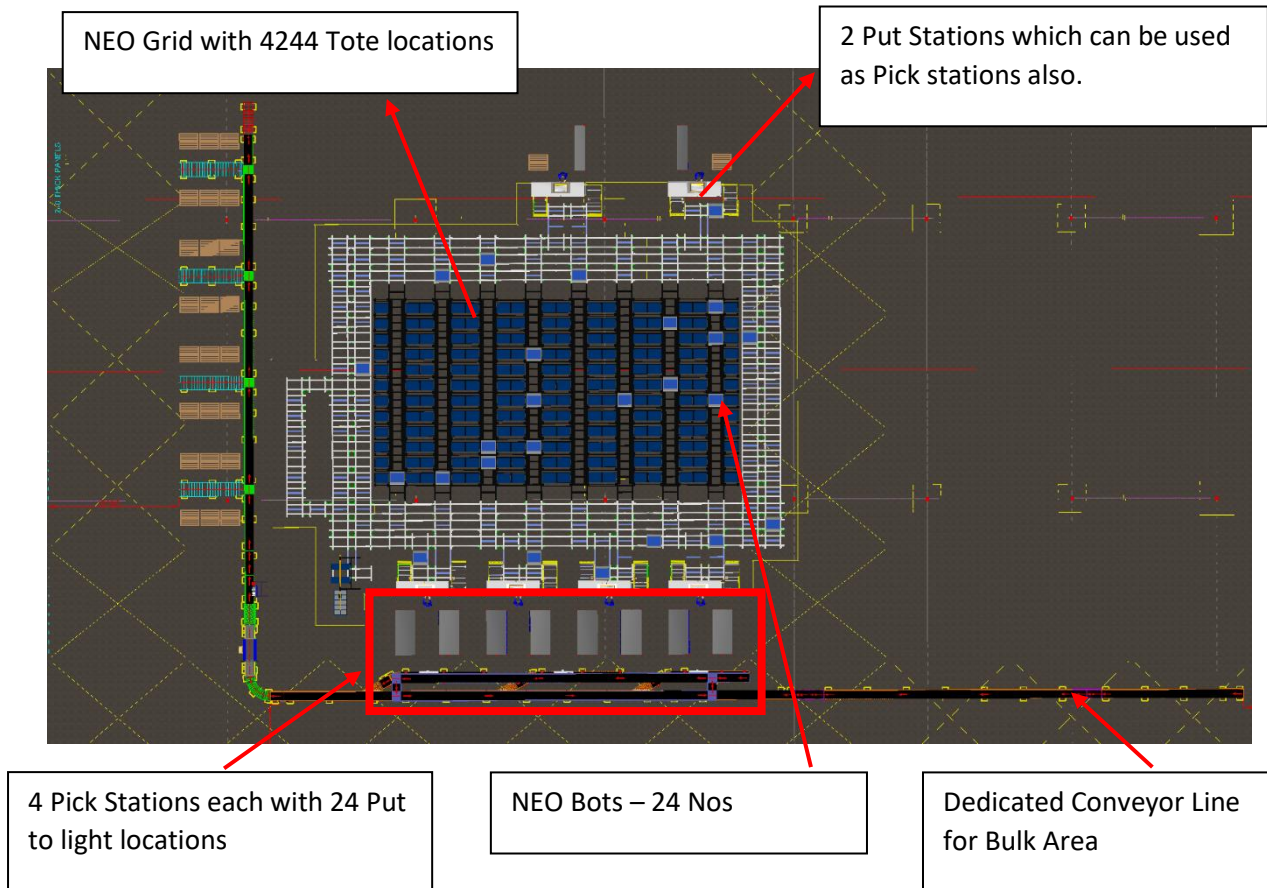
7.4 System Capacity at various areas

Activity	Throughput per hour
Pick Stations	96 Orders/Hour
Order Carton Takeaway Conveyor	900 Cartons/Hour
Automatic Label Printer	900 Cartons/Hour
Auto Sortation System	900 Cartons/Hour

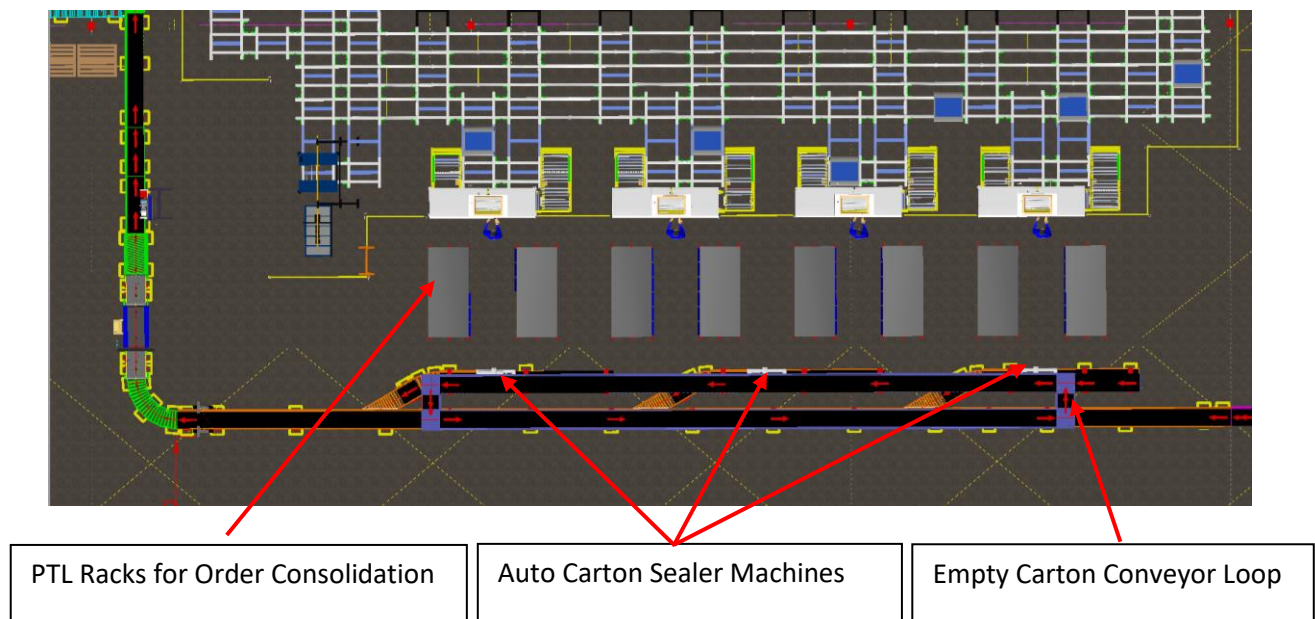
**All the above throughputs are calculated for average size Cartons/boxes. (300X300X300MM)*

7.5 Solution Layout Description:

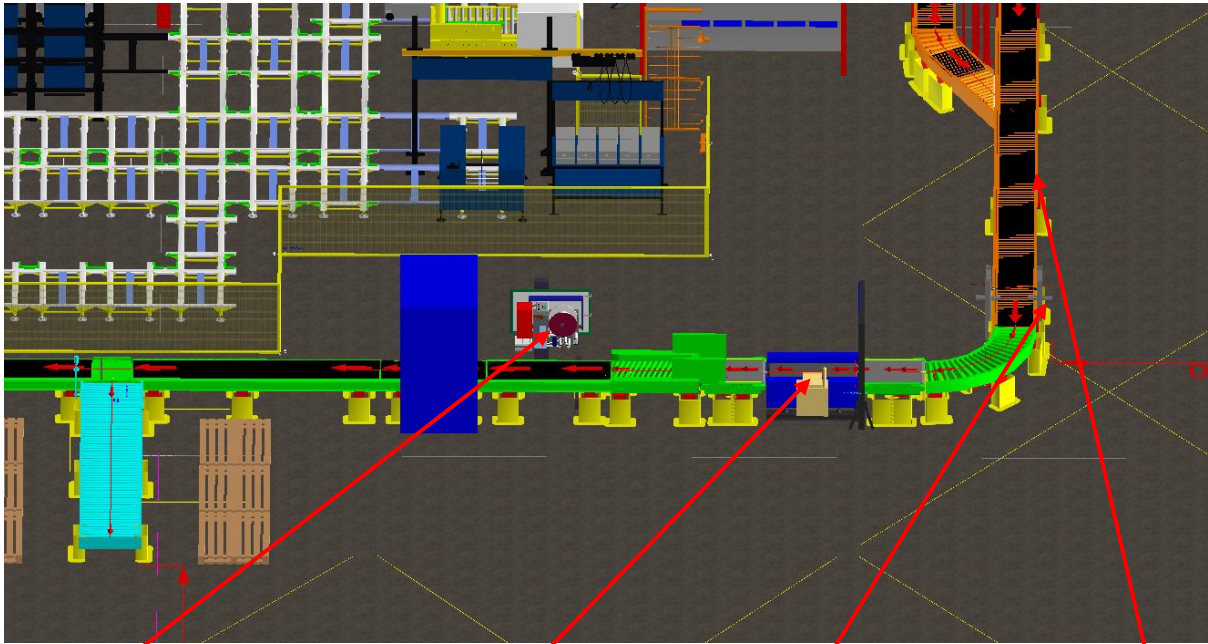
NEO Racking and Goods to Person Stations



Order Consolidation and Packing Area



Check weighing and Label Pasting Area \



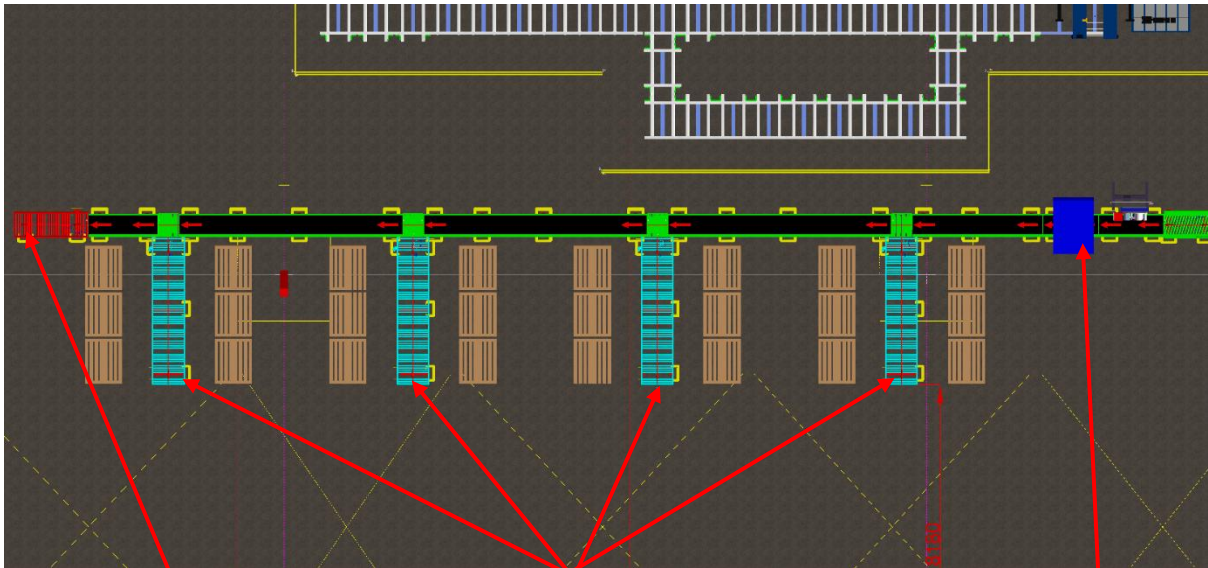
Automatic Label
Printer

Online Weighing
Conveyor

Side Barcode Scanners
for LPN Scanning

Highway conveyor for
order cartons

Route Wise Sortation Area



Rejection/Sort fail
Destination

Output Sorting Destinations – 4
Nos.

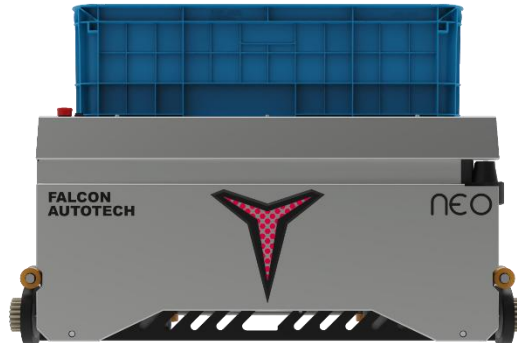
Top Side Scanner for AWB
Scanning

7.6 Process Flow:

- System consists of 4224 / 5544 tote locations where all the SKUs are stored in the order of their priority with high moving SKUs in the bottom and low moving on the top rows.
- Orders to be processed are received by the Neo software system which then smartly allocates the orders to the PTL lights provided at each pick station.
- The operators on the pick stations map the PTL barcodes with the carton barcodes before beginning a pick wave. This process marries the carton ID to the order ID.
- The operators then begin the picking process. The Neo bots bring the SKU tote to a picking station required to fulfil orders at each station. The operator scans the SKU barcode and the order PTLs that required the scanned SKU starts to glow. The PTL module also shows the quantity needed in that order.
- The operator picks the required quantities and put it in the order cartons. Once completed the Neo bot takes the carton away and another bot brings a new SKU carton to the station.
- In case the previous carton is required in any other pick station, it is taken there or else is taken back to storage.
- In case an order is completed, the PTL module displays a message and prints a packing slip. The operator puts the packing slip in the carton, fill the packing material and pushes the carton towards the back of the rack and confirms on PTL module. In case a carton is full, but the order is still not completed, the operator scans the close carton barcode on the PTL module, fill the packing material and pushes the carton back. He then places a new carton on the shelf and maps its barcode with open carton barcode on PTL module to update the system with 2nd carton for the same order.
- The process of pick and put continues at all stations simultaneously until the orders of that wave are completed.
- The operator (Runner) picks the completed carton from the rack, closes its flap and places it on the taping infeed conveyor.
- The cartons from the three taping lines merge into the highway conveyor. The cartons from the bulk area conveyor also merge into this highway conveyor.
- All the cartons on the highway conveyor are scanned for the barcode, weighed and a shipment label is generated and pasted on the cartons automatically.
- The shipment label is then scanned, and the cartons are sorted into one of the 4 sorting locations as per the sorting recipe.
- During inbound, the pallets are brought to put away stations. The SKU barcode is scanned, and eaches are placed into the NEO Tote. Put to light setup with 12 locations per put station is provided for buffering inbound neo totes.

8. System Components

8.1 NEO



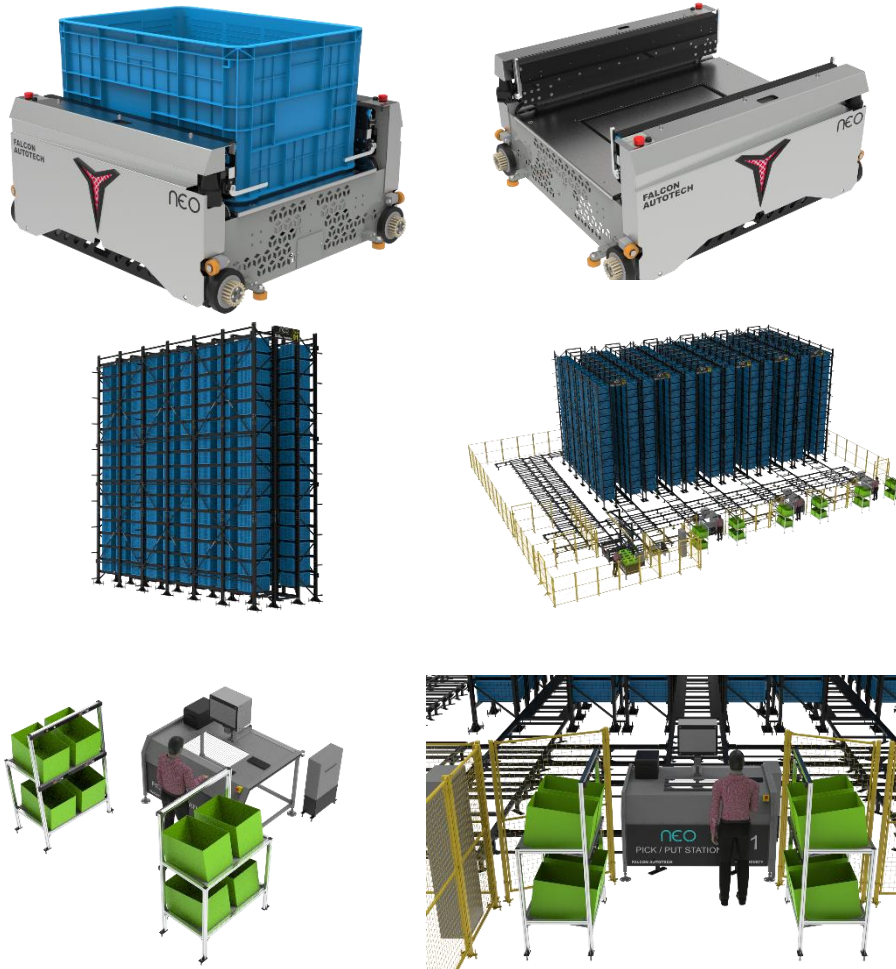
NEO is an end-to-end engineered robotic system for storage and order picking. It is an integrated system consisting of storage bins or totes, storage grids, robots, goods-to-person workstations, and software.

NEO provides higher storage density, is more adaptive, and can be deployed quickly in a cost-effective manner compared to traditional material handling systems. NEO moves flexibly within the storage grid using lidar and AI software capabilities to store and retrieve totes.

- NEO operates on all three axes within the grid to store totes at predetermined locations.
- When the wave starts NEO retrieves and brings the totes having desired products to the picking stations.
- Humans at the picking stations pick items into the order totes. Once the picking finishes NEO stores the inventory totes back to the grid.
- Post completion of orders the order totes can be routed to the packing stations using a series of conveyors.

Technical Specifications

						
Self Weight	Dimensions	Working Parameters	Performance	Movement	Safety	Fully customizable
200 Kgs Rated load without bin -40 kgs	BOT 1016*906*500mm External Bin Size 810*568*425 mm Bin volume – 5.8 cubic feet	0 to 35 Degrees Navigation mode – QR code	Works 24/7 manual battery swapping	Moves in 3 dimensions X, Y, and Z Maximum operating speed of 2m/s	360 degrees Obstacle detection Lidar in X, Y and Z axis (3D)	Bin Compartment – 2/3/4/6/8 Maximum storage height -12 m Max bins storage per tower -22



NEO Bots: The NEO Bots autonomously and independently move in all three dimensions under the structure and bring products to and from human or robotic picking operators. The robots move at 2 m/s and can carry bins weighing up to 40Kgs. NEO Bots are equipped with a swappable lithium battery which helps the system provide maximum uptime.

Racks: The NEO system is built around a storage grid made up of specialized racking structures that can reach a height of 60 feet to enable vertical space utilization in the warehouse. With zero sensors and electrical cables, the racking structure requires no maintenance.

Bins: Products are stored and moved in specialized bins or totes designed to be handled by NEO system. Each bin can be partitioned into 2/3/4/6/8 sub-sections for better product segregation.

Picking workstations: NEO system comes with operator workstations and PTLs for respective use cases or order picking and storage. The PTL station supports 8/12/16/24 parallel orders by maximizing the commonality of SKUs.

NEOIT: NEOIT is Falcon's in-house developed warehouse control software (WCS) that controls the NEO System. NEOIT does task planning, route planning and records the real-time positions of bins and NEO Bots.

TECHNICAL SPECIFICATIONS		
Basic Specifications	Specification Parameter	Parameter Description
	Model Designation	NEO
	Rated Load	40 Kg
	Self-Weight	200 Kg
	Position Accuracy	+ - 2 mm
	Dimensions (L*W*H)	1016 * 906 * 500 (mm)
	Ground Clearance	20 mm
	Navigation Mode	QR CODE
	External bin Size (L*W*H)	810 * 568 * 425 (mm)
	Bin Volume	5.75 Cubic Feet
	Bin Compartment	2 / 3 / 4 / 6
	Maximum Storage Height	14 Metres
	Maximum Bin Storage per tower	22 Levels
Performance	Driving Mode	Electric Servo Motors
	Control Mode	Manual / Automatic
	Movement	3-D (X, Y, Z Directions)
	Motion Signal	Light Prompt
	Works	24/7
Battery Specifications	Driving Voltage	DC48 V
	Battery Life	Upto 2500 Cycles @ 1C DOD 80%
	Endurance	6 to 8 Hours
	Charging time	Upto 2 Hours
	Charging Mode	Manual Battery Swapping
	Obstacle Detection	360 Degrees (LIDAR in X, Y, Z Axis)
	Emergency Stop	Front and rear emergency stop buttons
	Communication	Industrial Wi-Fi IEEE 802.11 a/b/g/n
	Pick / Put (PP) Station Safety	Light Curtain Sensors
	Working Area Safety	Doors equipped with Saftey Locks
Operating Parameter	Safety on Z - Axis Movement	Self Locking Brakes
	Global Saftey (for all Robots)	Using Global Emergency Stop at PP Stations
	Max Operating Speed (With load) in X & Y Axis	2 m/s
	Max Operating Speed (With load) in Z Axis	1 m/s
	Stopping angle Accuracy	+ - 0.5°
Special Features	Stopping position Accuracy	+ - 2 mm
	Pick / Put Guidance	Yes, through focus light
	Low voltage Battery alarm	Yes, robot will go to the charging station automatically in case of Low Voltage
	Pick / Put Stations Interchangeability	Yes, any station can be used as Pick / Put Station
	Handling fast moving, non-fast moving SKU's to locations near to stations	The system Dyanmically stores the fast- moving SKU's closer to the pick stations and the slow movers farther inside the grid. This enables shorter trip time of the bots thereby greatly reducing the number of robots required.
	FIFO option for the SKU's	It enables the user to consume the SKU which were replenished basis their PUT timestamp.
	Bin recall Feature	It helps in audititng the SKU's
	Re-put Away Feature	It consolidates the less quantity SKU's which are kept in different locations in a single bin and frees the space for next SKU replenishment.
	Realtime Inventory Tracking	Real Time Inventory on the Dashboard
	Ease of maintenance	Mobile Tablet based control for maintenance
	Data Analytics	Detailed and Customized data extraction is possible
	Pick / Put Assist	Dashboards at each PP station that assists operator in pick / put to reduce to reduce the chances of error on the station.

8.2 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- It comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	600
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Load capacity/unit length of Conveyor	kg/m	30
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	100
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

8.3 PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frames.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	600
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Load capacity per unit length of Conveyor	kg/m	30
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew
Drive	make	Lenze/Seimens/Omron/Allen Bradley
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

8.4 Pop- Up Unit

Falcon's Belt Popup device is used for 90° transfer of products.

Some Silent Features of Falcon's Pop-Up:

- Low Noise
- Maximum uptime.
- Electric based popup.
- Minimal maintenance
- High safety standards.



Specification	UOM	Remark
Carton Box cross-transfers (Pop-Up Units)	Nos	4
Manufacturer Name	Name	Falcon/Pulse/Itoh Denki
Product Dimensions	mm	Max Dimensions: 470 X 315 X 285 Min Dimensions: 270 X 220 X 160
Type – Equipment	Type	90 Degree Divert
Type – Operation	Type	Sortation
Mechanism for Pop-Up	Type	Electric
Drive Power Rating (Lifting)	Watt	40
Type of Motor	Type	DC
Drive Power Rating (For Rollers)	Watt	40
Type of Motor	Type	DC
Ingress Protection type	Type	IP 54
Provision for Carton Box / Carton Detection Sensors	Yes/No	Yes
Load Capacity of Carton Box	Kg	20
Sorting Capacity	PPH	900

8.5 Pick/Put to Light Module

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with Client's WMS/ERP Systems.

The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keeps on altering the PTL configuration to increase the system productivity and give you an extra competitive edge.

The System involves a collection of physical sorting locations representing either a customer order/address location or any other logical sorting group. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.

It is a paperless technology that provide high throughput with single operator with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Dimensions	Mm	Around 165 X 50
PTL location	Nos	As per Layout requirement
Weight of the carton box	Kg	25
LED Display	Type	3 Digit Alpha numeric Rotating Display
LED Size	Mm	~ 25 X 20
LED Indicator	yes/no	YES
Acknowledgment Button	yes/no	YES
Function keys	yes/no	4 Keys
Power Requirements	Type	3 Phase UPS Power
Communication	Type	TCP/IP and Serial
Operating temperature	Deg.	45 Degrees
Software Interface	Type	Over APIs or other methods
Type of mount	Type	Clip On Type on Rails

8.6 Automatic Barcode Scanners

The system is integrated with automated inline barcodes scanning capable for scanning 1D and 2D barcodes with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Sick/Cognex
Barcode Type	Type	1D/2D
Module Size	Mils	1D >= 15
Product Dimensions	Mm	Max Dimensions: 470 X 315 X 285 Min Dimensions: 270 X 220 X 160
Number of Codes	Nos	1
Location	Side	LPN Scanning – Side Scanning AWB Scanning – Top Side Scanning
Orientation	Type	Unidirectional
Colour of bars	Colour	Black
Under Foil	yes/no	No
Code Length	Mm	50
Code Height	Mm	15

8.7 Hand-Held Barcode Scanners

The system is equipped with handheld barcode scanners capable of scanning 1D/2D barcodes for backup.



8.8 Weighing Conveyor

A Weighing Conveyor, also known as a Weigh Belt Conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.



Specification	UOM	Remark
Manufacturer Name	Name	Bizerba/MT/Similar
Product Size	Mm	Max Dimensions: 470 X 315 X 285 Min Dimensions: 270 X 220 X 160
Minimum Weight	Kg	0.2
Maximum Weight	Kg	20
Weighing Accuracy	Kg	+/- 0.06 or 0.5% of product weight, whichever is higher
Weighing Resolution	Kg	0.001

8.9 Automatic Label Applicator

The system is integrated with an automated label applicator that uses pneumatic applicator for pasting. The printer can print and paste on varying box sizes on the top side.



8.10 Automatic Taping Machine

The system is integrated with an automated carton sealing machine. The machine auto adjusts to the size of the box within permissible range and tapes from top and bottom. The operator will need to close the flaps manually before pushing the box into the machine.



8.11 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. More Storage in less floor space.
2. Utilizing vertical height up to 11 Mete.
3. High operational throughput.
4. Improved Productivity.
5. Designed for minimal upkeep, ensuring consistent reliability and ease of support.
6. Life Cycle Value.
7. Modular in nature allowing for an easy expansion.

9. Proposed System Technical Details

9.1 Mechanical equipment

Pos.	Qty.	Description	Value
FM1	1	NEO Automation - Neo Racking and Structure - Neo Pick Stations - Neo Put Stations - Neo Bots - Infra for Neo - PTL Push Back Racks (6X3 Type) - PTL Standard Frames (6X3 Type) - PTL Modules with Ducts – Pick Stations - PTL Modules with Ducts – Put Stations - PTL Control Boxes - Hand-Held Barcode Scanners - Fencing	4224 Bin Locations 4 Nos. 2 Nos. 24 Nos. 1 Set. 8 Nos. 2 Nos. 96 Nos. 24 Nos. 6 Nos. 6 Nos. 1 Set.
	1	Conveyors Automation and Sortation System - PVC Belt Conveyor - Gravity Roller Conveyor - Accumulation Roller Conveyor - Powered Roller Conveyor - Roller Merge (30°) - Roller Turn (30°) - Roller Turn (90°) - Pop-Up Diverts for Recirculation Loop - Automatic Carton Sealer - Automatic Side Barcode Scanner (1D) - Weighing Conveyor - Aligner - Automatic Label Printer - Top Side Scanning Tunnel (1D/2D) - Pop-Up Diverts for Sortation	2 Nos. 18 Meter 79 Meter 26 Meter 3 Nos. 3 Nos. 1 Nos. 4 Nos. 3 Nos. 2 Nos. 1 Nos. 1 Nos. 1 Nos. 4 Nos.
FA1	1	Bulk Line Infeed Conveyor Accumulation Roller Conveyor	34 Meter
FA2	1	Sliding Roller Gate Conveyor Sliding Roller Gate Conveyor	1 Nos.

9.2 Electrical Equipment

Electricals			
FE1	1	Consists of • Main Power Distribution panel • Main Control Panel • Sorter Drive Panels • Network Switches • Field Cabling	Included

9.3 Control System

Components			
FC1	1	Control System Falcon Software and Software Support Integration with WMS HAA Server PC and Racks	Included

9.4 Control System

Components			
FS1	1	Project Management A Senior project manager oversees the project till completion. Entire coordination between different teams at client and Falcon end is managed by the project manager stationed in India.	Included

9.5 Control System

Components			
FS2	1	Installation and Commissioning The Installation and commissioning are carried out by specialists of supplier based on an agreed lump sum – for a normal working time of 8 hours per day and 5 days per week. ○ Board & Lodging ○ Travel Expenses ○ Insurance ○ Tools	Included

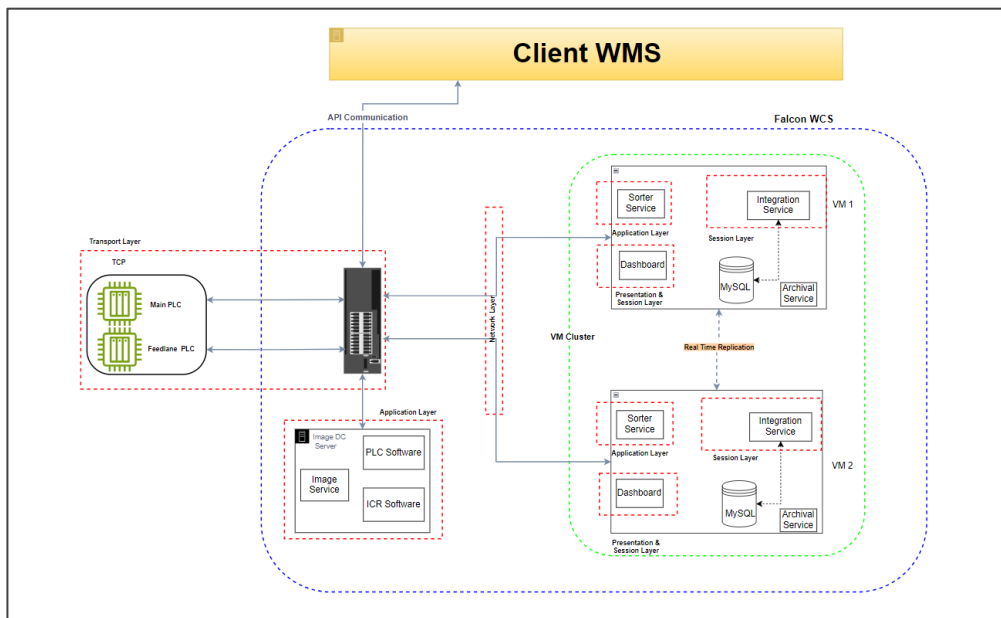
10. Falcon's WCS CONTROLIT

Falcon WCS (Warehouse Control System) is an in-house developed IT solution by Falcon Autotech, serving as the brain behind the company's sortation solutions. It manages the real-time movement of goods and data across the system, ensuring efficient operations in high-throughput warehouses. Falcon WCS integrates seamlessly with Warehouse Management Systems (WMS), Transport Management Systems (TMS), and other external applications via APIs to enhance operational efficiency.

10.1 System Architecture

High-Level Design (HLD) Overview

The Falcon WCS integrates with external systems like the Warehouse Management System (WMS) and Transport Management System (TMS). Communication occurs via **APIs/WSDL/MQ Communication Protocol etc.** ensuring smooth data flow for order management, shipment tracking, and other critical operations.



Key Components:

- **Presentation and Session Layer:**
 - **MySQL Database:** Stores operational data, shipment details, and sortation instructions.
 - **Sorter Services:** Responsible for managing sorting logic and directing parcels to appropriate destinations.
 - **Dashboard:** Provides a user interface for real-time monitoring of warehouse operations and performance metrics.
 - **Integration Services:** Handles communication with external systems (e.g., WMS, TMS) and ensures data consistency across platforms.

- **Application Layer:**

- Image Services: Processes and manages images captured during the sortation process.
- ICR Software: Utilizes Image Character Recognition to read parcel labels and identify shipment information.
- PLC Software: Interfaces with Programmable Logic Controllers to manage the physical movement of parcels and control sortation equipment.

- **Transport Layer:**

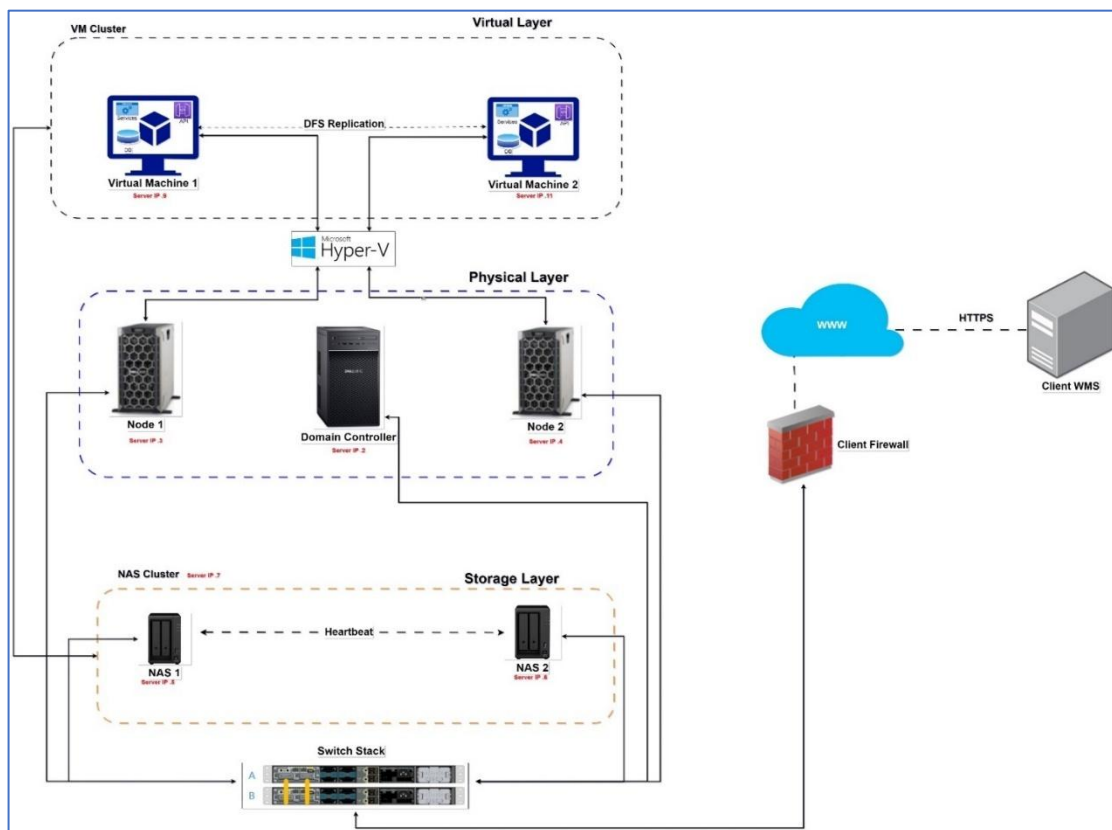
- Sorter PLCs: Receive commands from the session layer (Sorter Services) and execute sorting operations based on real-time data.

- **System Communication:**

All layers are connected via a Stacked Switch, which provides internet and intranet connectivity. Communication between the sortation system and external systems for results or shipment data occurs through this switch.

10.2 High Availability Architecture

The Falcon WCS architecture ensures uninterrupted operations using a **High Availability (HA) Server setup**. The system is designed to handle both planned and unplanned downtime, providing robust mechanisms for failover, replication, and data redundancy.



Key Components and Features of the High Availability Architecture:

- **Stacked Switch:**
 - Centralizes data exchange between **NAS, Nodes, Domain Controller (DC)**, and peripherals.
 - Analyses packet headers to reduce unnecessary data transmission, enhancing LAN efficiency.
- **Domain Controller:**
 - **Heartbeat Monitoring:** Tracks the status of nodes and initiates VM failover when necessary.
 - **Image Hosting:** Stores and manages images received from the ICR (Image Character Recognition).
- **NAS (Network Attached Storage):**
 - Centralized data storage providing access to connected devices and Virtual Machines.
 - **Redundancy:** Two NAS boxes with mirrored drives ensure data protection and availability, offering a failsafe against hardware failure.
- **Node:**
 - **Hyper Terminals:** Nodes host and manage Virtual Machines (VMs) to run the warehouse control systems and related applications.
 - **Clustering:** Nodes are clustered using **Microsoft Windows Cluster** to enable failover protection, ensuring continuous operation even in case of hardware failure.
- **Virtual Machine & InnoDB Cluster:**
 - **Primary VM:** Hosts Falcon WCS services, while a secondary backup on the node ensures failover through **Network Load Balancing (NLB)**.
 - **InnoDB Cluster:** Ensures data replication using a **Master-Slave-Slave** setup for MySQL databases, maintaining consistency and availability.
- **NAS Cluster:**
 - **Unified File System:** NAS nodes share files across the cluster, ensuring no data loss during failover or disaster recovery.
 - **Backup NAS:** Provides redundancy by replicating data between two NAS boxes, further safeguarding against failures.

Disaster Handling:

- **Recovery Time Objective (RTO) & Data Loss Objective (RPO):**
 - **VM Cluster Failure:** RTO = 1 hour; RPO = 1 hour.
 - **Node Failure:** No impact with a single failure; RTO = 4 hours if both nodes fail.
 - **NAS Failure:** Backup NAS available with no downtime, ensuring continued operation.

10.3 WCS User Interface

Overview

The Falcon WCS features a robust, user-friendly **Dashboard** that provides real-time visibility into warehouse and sortation operations. The Dashboard serves as the primary interface for monitoring key system metrics, tracking performance, and ensuring smooth operations.

Dashboard Overview

The WCS Dashboard offers real-time data visualization, helping warehouse operators and IT teams make data-driven decisions. Users can monitor system health, performance, and detect anomalies through an intuitive graphical interface.

Key Features of the Dashboard:

- **System Health Monitoring:** Displays metrics such as CPU utilization, memory usage, disk performance, and system load across the infrastructure.
- **Real-Time Sortation Monitoring:** Shows the real-time movement of parcels within the sortation system, including chute assignments and shipment statuses.
- **Error Reporting:** Notifies users of system errors, network disruptions, and potential failures in real-time, allowing for quick resolution and minimal downtime.
- **Performance Metrics:** Provides detailed reports on sortation throughput, parcel handling times, and system efficiency to ensure that warehouse targets are met.
- **User Role Management:** The dashboard allows different levels of access based on user roles, ensuring that the right personnel can view or manage the system as needed.

In the context of this IT dashboard, the following user interactive screens are provided

- **Dashboard (Home Screen):** Provides an overview of important metrics, data visualizations, and summary information related to the IT system or processes.



Figure 1 Dashboard

- **Live Bags:** Displays real-time information and status updates regarding bags or parcels currently in transit or being processed.

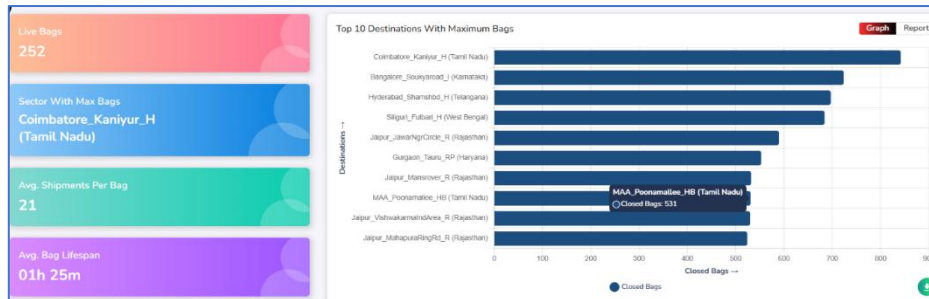


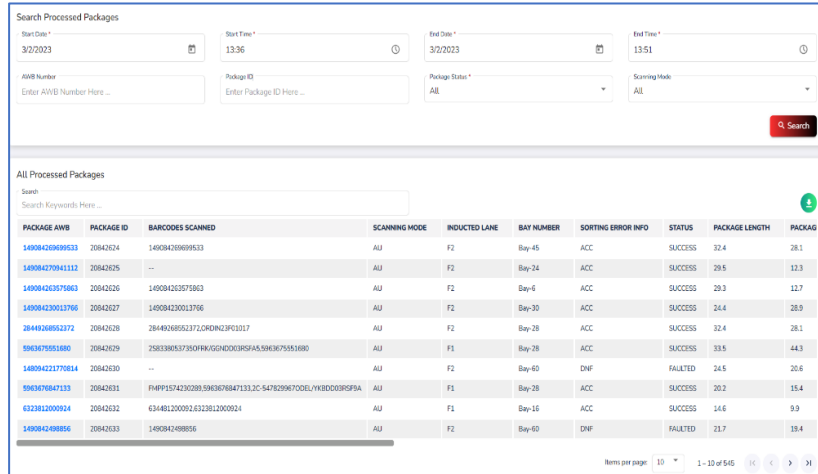
Figure 2 Live Bags

- **Bay Status:** Offers insights into the status and availability of different processing bays or areas within the system.

Bay Status				
Search Search Keywords Here...				
BAY ID	BAY STATUS	BAY TYPE	BAGGING MODE	SHIPMENTS SORTED IN BAY
Bay-1	ACTIVE	REGULAR	DIRECT	0
Bay-2	ACTIVE	REGULAR	DIRECT	0
Bay-3	ACTIVE	REGULAR	DIRECT	0
Bay-4	ACTIVE	REGULAR	DIRECT	0
Bay-5	ACTIVE	REGULAR	PTL	19
Bay-6	ACTIVE	REGULAR	DIRECT	0
Bay-7	ACTIVE	REGULAR	PTL	27
Bay-8	ACTIVE	REGULAR	PTL	7
Bay-9	ACTIVE	REGULAR	PTL	4
Bay-10	ACTIVE	UNMAPPED	DIRECT	0

Figure 3 Bay Status

- **Processed Packages:** Shows details and statistics related to packages or items that have been successfully processed or handled by the system.



PACKAGE AWB	PACKAGE ID	BARCODES SCANNED	SCANNING MODE	INDUCTED LANE	BAY NUMBER	SORTING ERROR INFO	STATUS	PACKAGE LENGTH	PACKAGE
14008420809533	20842624	14008420809533	AU	F2	Bay-45	ACC	SUCCESS	32.4	28.1
14008420809533	20842625	---	AU	F2	Bay-34	ACC	SUCCESS	29.5	13.3
14008420809533	20842626	14008420809533	AU	F2	Bay-6	ACC	SUCCESS	29.3	12.7
14008420809533	20842627	14008420809533	AU	F2	Bay-30	ACC	SUCCESS	24.4	28.9
2844920809533	20842628	2844920809533	AU	F2	Bay-28	ACC	SUCCESS	32.4	28.1
14008420809533	20842629	14008420809533	AU	F1	Bay-28	ACC	SUCCESS	33.5	44.3
14008420809533	20842630	---	AU	F2	Bay-60	DNF	FAULTED	24.5	20.6
14008420809533	20842631	14008420809533	AU	F1	Bay-28	ACC	SUCCESS	20.2	15.4
14008420809533	20842632	14008420809533	AU	F1	Bay-16	ACC	SUCCESS	14.6	9.9
14008420809533	20842633	14008420809533	AU	F2	Bay-60	DNF	FAULTED	21.7	19.4

Figure 4 Processed Package

- **Configuration Setting:** Enables users to configure and customize various settings and parameters within the IT system or dashboard.

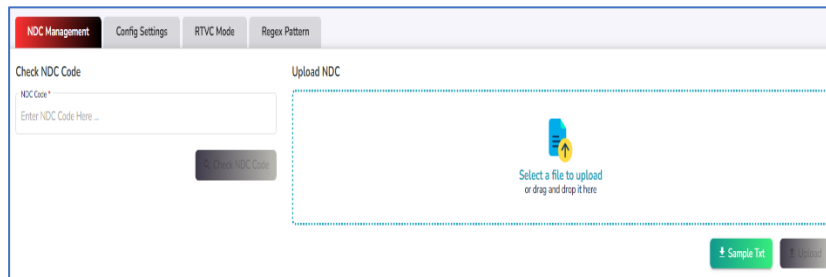


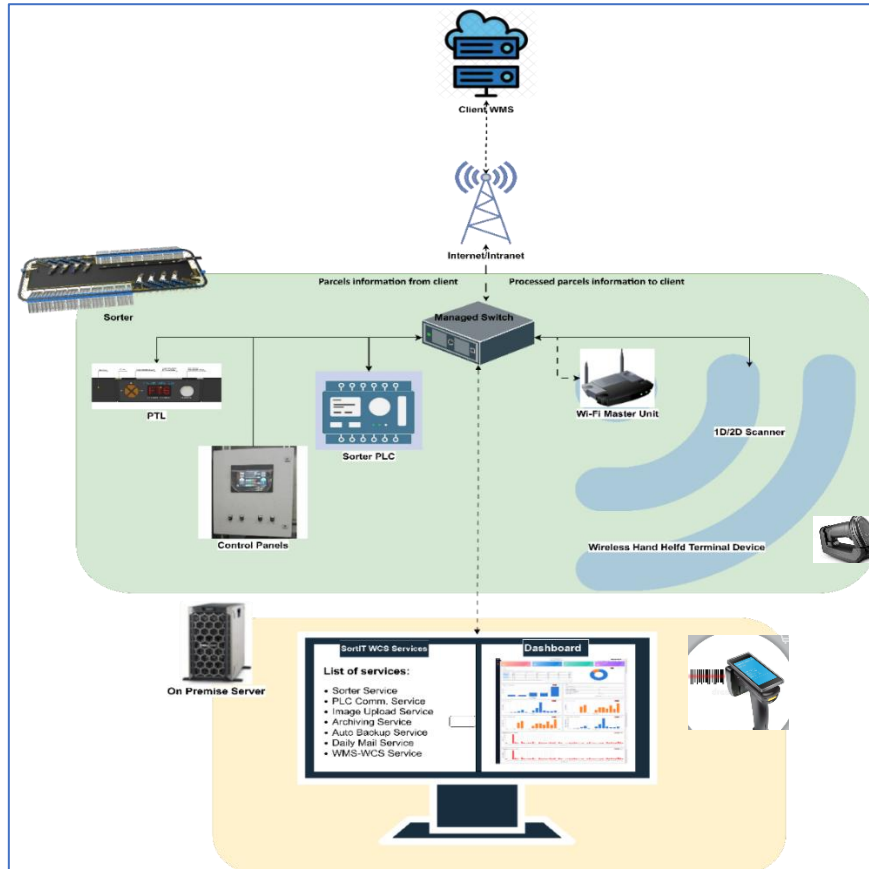
Figure 5 Configuration Settings

- **Report & Analysis:** Allows users to generate and access comprehensive reports, analytics, and insights based on the data collected by the IT dashboard.
- **Rejection Bay Mapping:** Provides functionality to map and manage rejection bays or areas where packages are deemed unsuitable for processing.
- **Alarms:** Displays alerts, notifications, or alarms related to system events, errors, or anomalies that require attention or investigation.
- **Calibration Settings:** Allows users to adjust and calibrate system settings, parameters, or sensors to ensure accurate and reliable performance.
- **Operator Management:** Offers features and tools to manage and monitor the operators or personnel responsible for operating the IT system.
- **User Management:** Provides functionality to manage user accounts, permissions, roles, and access levels within the IT dashboard.
- **User Guide:** The "User Guide" page offers comprehensive documentation and instructions on how to use the IT dashboard effectively. It serves as a reference guide for users.

10.4 Communication Architecture

Overview

Falcon WCS operates within a highly interconnected system, ensuring seamless communication between the WCS server, on-premises devices (such as Sorter PLCs, PTL devices, 1D scanners, and HHT devices), and client systems. This communication architecture facilitates real-time data exchange and operational control, optimizing sortation processes and warehouse efficiency.



On-Premises Communication

Sorter PLC Devices:

- **Protocol:** Falcon WCS communicates with Sorter PLCs using either the **Siemens S7** protocol or the **Omron communication** protocol.
- **Functionality:** The Sorter PLC devices receive sortation instructions from the WCS and execute the sorting process by directing parcels to the appropriate chute based on the system's real-time data.

PTL (Pick-to-Light) Devices:

- **Protocol:** PTL devices communicate with Falcon WCS using the **TCP/IP protocol**.
- **Functionality:** The system sends commands to the PTL devices for guiding manual picking operations by lighting up indicators at the appropriate bins or shelves, improving operational accuracy and speed.

1D Scanners:

- **Protocol:** These barcode scanners also use the **TCP/IP protocol** to communicate with the WCS.
- **Functionality:** The scanners capture barcode data from the parcels, and this information is sent to the WCS for processing, such as determining sorting destinations.

HHT (Handheld Terminal) Devices:

- **Protocol:** The wireless **HHT devices** communicate with Falcon WCS over **Wi-Fi**.
- **Functionality:**
 - The HHT devices send scan input data (e.g., barcodes) to the server over Wi-Fi.
 - The WCS processes this data and sends the required output instructions back to the HHT device and associated PTL devices.
 - The HHT device executes these instructions, facilitating real-time decision-making and execution for operators.

10.5 Client Communication**Data Transfer Methods:**

- **API:** Falcon WCS can communicate processed data to client systems through **API** calls, allowing for seamless integration with external software.
- **MQ (Message Queuing):** Falcon WCS can also send data via **message queues**, ensuring reliable delivery of messages even during network downtime.
- **WSDL/XML:** For structured data exchanges, Falcon WCS supports **WSDL** and **XML** formats for client communication.
- **Other Protocols:** Additional methods for data transfer may include customized protocols depending on client requirements.

Purpose:

- The data sent to the client can include sortation results, system performance reports, and operational analytics, which can be used for further processing or reporting within external systems like **Warehouse Management Systems (WMS)** and **Transport Management Systems (TMS)**.

11.Key Components Make

Items	Make
Belts	Forbo/ Derco
Rollers	Falcon
Pop-up Units	Falcon
Volume Scanners	Cognex
Weighing Scales	Bizerba/ Mettler Toledo
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens/Omron
Control Panels	Rittal/ BCH
VFDs	Siemens/Lenze/ AB/ Omron
Cables	LAPP/ Equivalent
Switch Gear	Schenider/Equivalent
Bearings	NTN/SKF
Power Transmission Systems	Vahle
HMI's	Siemens

12.Principle of Safety

E-stops, pull cords and safety fencing will be provided in the system as per Falcon Standards.

12.1 E-Stops



12.2 Pull Cords Switch-



13. Infrastructure

13.1 Fire Protection-

Falcon's scope does not cover the design or provision of fire protection infrastructure, utilities, or related services. It is expected that the customer's sprinkler contractor will design and supply the in-rack sprinkler systems, including connectors and mounting brackets. These designs should be submitted to Falcon for review during the engineering phase.

Falcon will work in coordination with the chosen sprinkler supplier to determine the appropriate locations for the sprinklers and brackets. Modifications may be required to meet local fire safety regulations, which could influence storage capacity, timelines, and costs. Any additional sprinkler systems that might affect the overall design must also undergo review.

13.2 Power Supply-

The Customer must provide temporary power for installation and permanent power for commissioning. Protected multi-gang power points for workstations and peripherals will be supplied by the Customer, with planning for their locations done with the operations and IT teams.

13.3 Floor Requirements-

The Customer must provide flooring with appropriate loading strength and space at the site. Falcon assumes that the floor slab will not contain corrosive materials that could affect standard fixings.

13.4 Estimated Floor Load-

Estimated floor loads, including distributed and point loads, will be provided during the detailed engineering phase of the project.

13.5 Staging, Laydown and Assembly Area-

The Customer is required to provide sufficient space on the same floor, adjacent to the installation site, for staging, storage, and equipment assembly. If such space is unavailable, offsite locations or areas on different floors may be utilized; however, any related costs, including additional handling or transportation, will be the Customer's responsibility. This may also result in adjustments to the project timeline. Falcon will specify the necessary requirements for these areas during the design phase as part of overall project planning and coordination.

13.6 Site Access and Unloading-

The Customer is required to allocate sufficient on-site space for parking and staging shipping containers to facilitate Falcon's delivery schedule.

Additionally, the Customer is responsible for ensuring proper access to the building for equipment unloading, including providing enough functional dock levellers on all floors during installation. It is expected that access to the site and installation areas will remain unobstructed and available around the clock, as needed, to support project operations.

13.7 Lightning-

All lighting is excluded from Falcon's scope of supply and must be provided by the Customer or their contractor. This includes lighting for service areas, operational areas, and beneath platforms and walkways.

14. Program Organisation

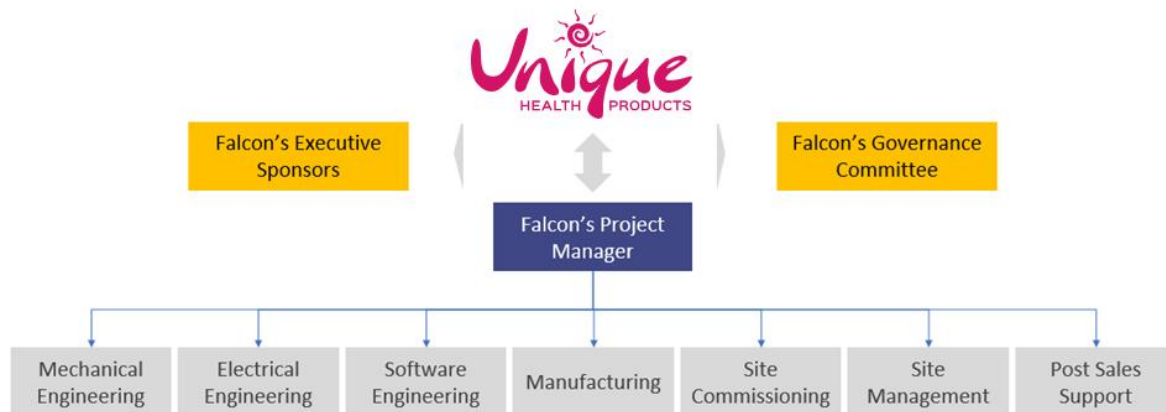
14.1 Project Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Weekly/Fortnightly Meeting UHP to share the Project Status.
3. Scheduling of the resource management.
4. Management of risks and opportunities.
5. Management of the list of anomalies or reservations.

14.2 Project Team

Team of 3 to 4 member from Projects team will co-ordinate on regular basis with UHP and Internal stakeholders for smooth execution of the project.



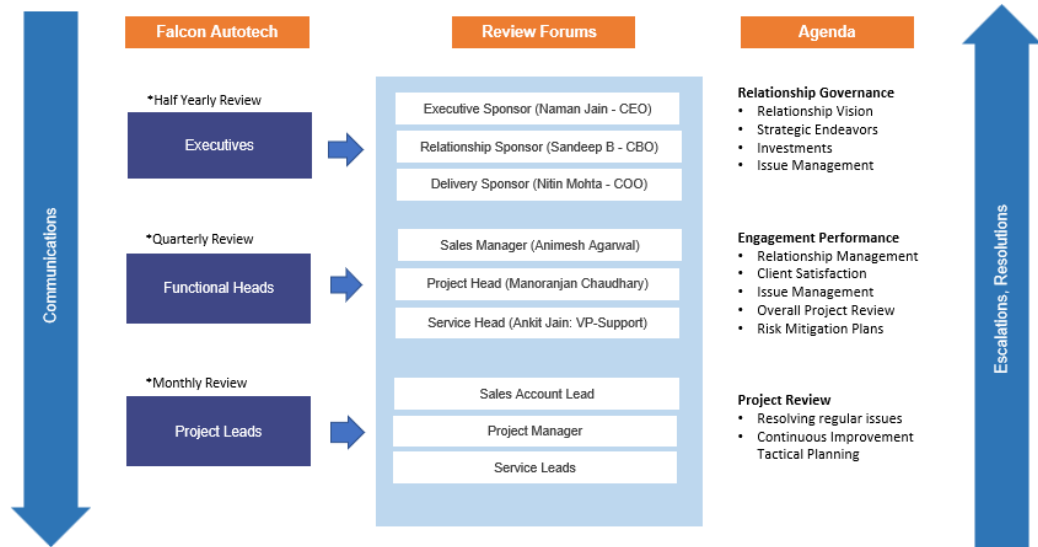
Falcon Project Manager is overseen and supported by Executive Sponsors. The Project Manager leads a multi-disciplinary team of members from all Falcon departments necessary to deliver the project, as shown in the diagram above.

14.3 Program and Account Management

The Project will be closely monitored under Falcon's Governance model as structured below.

Governance Model

Building strong customer relationship foundations on trust, transparency & communication



14.4 Program/ Project Schedule

Milestone	Stage	Starting Week	End Week
1	PO and Advance Receipt	Week 0	
2	Detail Design Approval	Week 2	Week 6
3	Manufacturing & Dispatch	Week 6	Week 30
4	Material receipt on site	Week 46	Week 37
5	Installation and Commissioning	Week 37	Week 45

**Above plan is tentative, detailed plan will be shared post order confirmation.*

15.UHP's Responsibility

15.1 UHP's Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- UHP to provide HPT/ Forklift for material movement for complete duration.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- The provision, during the commissioning phase, of the power supply necessary for the operation of the shipmentsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of

FALCON need.

- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area(s) to be protected against direct weather conditions.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installation site) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

15.2 Responsibilities of UHP During the Tests

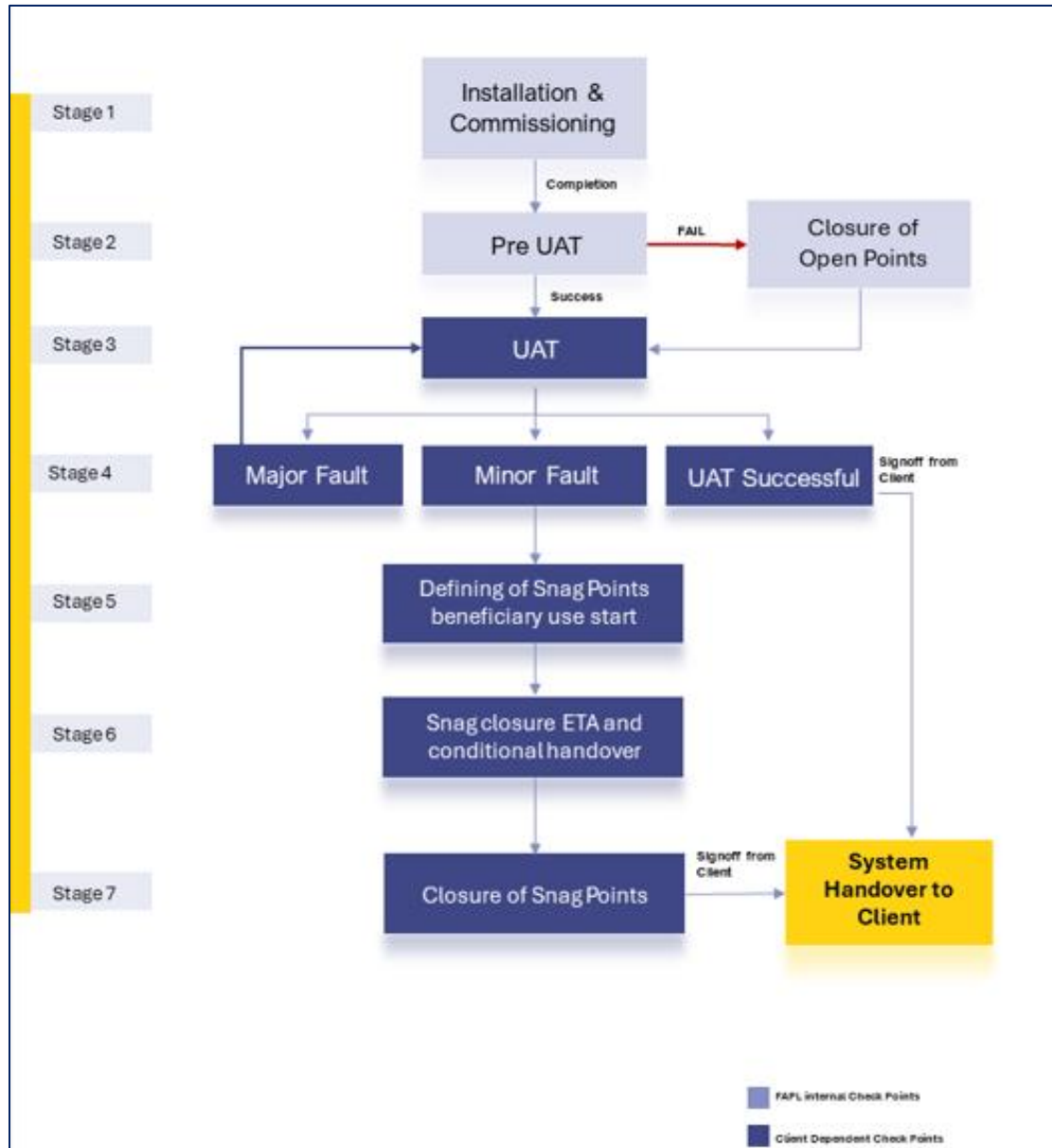
- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the shipments correctly.
- Verify with Falcon the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm test results with Falcon.

15.3 UHP's Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough space for desks or tables and chairs for the trainer and trained staff.
- Check the prerequisites of the people who must follow the training, e.g. the qualifications of the technical staff.
- The invitation of staff to attend the training courses will be at the expense of UHP.

16. System Testing

The system handover will be following the shown below workflow. Each of the stages are defined in the document below.



16.1 Installation and Commissioning

Completion of all the activities to get the system up and running.

16.2 Pre-UAT

Pre-UAT (User Acceptance Testing) involves a series of preparatory steps and tests conducted before the formal UAT phase begins. This stage ensures that the system is ready for end-user testing and meets the necessary requirements and standards. Below are some key checks of Pre UAT:

- **Physical Verification:** Ensure all functional and non-functional requirements are clearly defined and understood. Verify that all the necessary specifications for the sorting system are documented.
- **Integration Testing:** Conduct integration tests to ensure all components (conveyors, scanners, sorters, control systems, etc.) work together seamlessly. Validate the interfaces between the sorting system and other warehouse management systems (WMS) and databases.
- **Performance Testing:** Test the system's performance with some dummy product to ensure it can handle the expected volume of parcels or items. This test also done to identify and rectify any performance bottlenecks or inefficiencies.
- **Functional Testing:** Execute test cases to verify that the system performs all required functions correctly. Include scenarios for different types of products, error handling, and edge cases.

By thoroughly conducting pre-UAT activities, Falcon team will ensure that the system is stable, efficient, and ready for the formal UAT phase, where end-users can validate its functionality and performance in a real-world scenario.

16.3 UAT (User Acceptance Test)

User Acceptance Testing (UAT) for the System is the final phase in the implementation process where the end users test the system to ensure it meets their requirements and works as expected in a real-world scenario

Falcon will provide the UAT Plan to Client prior to the commencement of any tests. The UAT will include the following:

- Test prerequisites.
- Daily plan/schedule
- Personnel responsibilities
- Dashboard data and test loads required.
- Specific test procedures
- Expected test results.

User Acceptance Test Parameters

Test parameters will be discussed during the DAP phase and will be a part of the BRD.

Note: In conducting this evaluation/UAT, Falcon team adhere to its Standard Operating Procedure

16.4 Minor & Major Faults

Any faults encountered during Testing will be categorized by Falcon as either Minor or Major faults as defined below.

- **Minor Faults** are defined as those affecting a limited area or single component that has no impact on the test. Minor Faults will be added to the System Snag point list and will not inhibit the continuation of testing.
- **Major Faults** are defined as those having impact that result in the inability to demonstrate the subject functionality. Major Faults may require re-starting of the test.

16.5 System Snag Point

Once the UAT completed, all the minor faults will be considered as snag points. Falcon team will publish Snag Point List information to the client.

System Snag Point list information will include the following:

1. Issue.
2. Date identified.
3. Area and/or unit.
4. Category (mechanical, electrical, or software).
5. Remedy responsibility (responsible person[s] or organization[s]).
6. Target completion date.
7. Snag point completion verification and signoff.

Once the snag point list information's shared with client, client can start the beneficiary use of system and Client needs to sign off on Start of System Warranty and conditional handover.

16.5.1 Practice for Snag Point Sign Off:

1. Falcon personnel will maintain and distribute the System Sang List throughout UAT.
2. As each issue is corrected, Client will verify the resolution and provide sign-off for that issue on the System Snag Point List.

16.5.2 System Handover Letter

After successful completion of Acceptance Testing or closure of all snags, Falcon will furnish Client a letter, which will address the following:

- The subject System has been installed and accepted by Client.
- The final payment amount that is required and the designated due date.

17. Training

Training for the recently installed material handling system at UHP's facility is an integral part of the implementation process. Falcon employs a business-linked learning approach in the design, development, delivery, and evaluation of its training programs.

Falcon adheres to a well-defined process tailored to create learning content specific to UHP's installation and application of their systems.

17.1 Operations Training

System Operations Training is provided to offer participants an understanding of their area from an operational perspective. Falcon targets critical operations in the assigned functional area, covering flow control, system operation, and control devices. Discussions also emphasize the impact of individual and area performance on upstream and downstream workers.

17.2 Maintenance Training

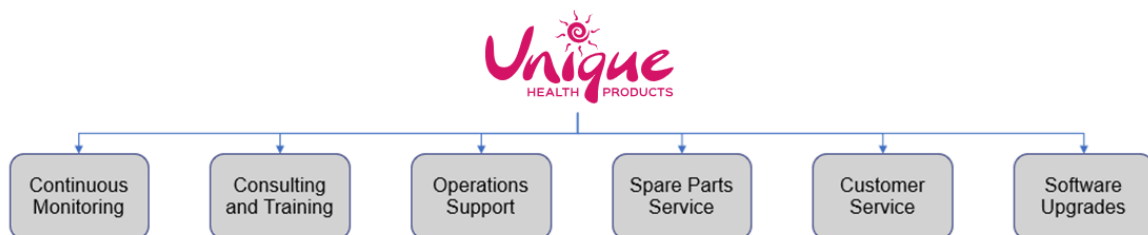
The Maintenance Training curriculum encompasses mechanical, electrical, and controls aspects of the installed system. Topics include safety, system construction and installation, system operation, maintenance and repair procedures, and technical documentation.

Falcon recommends assigning personnel responsible for system maintenance to work with the installation and commissioning crews during their final weeks on-site. Formal training courses for maintenance personnel will be conducted before system startup, featuring classroom lectures, audio-visual presentations, and hands-on demonstrations with the installed system. Site tours will be organized to highlight common operational issues affecting system uptime.

17.3 Information System Training

Information System training covers the computer hardware and software aspects of the Automated Material Handling System.

18. Service Support



Benefits of Falcon's Service and Support Plan:

- 24X7 Online Support from Falcon's global central team
- Critical Spares are supplied and retained on-site to ensure continuity and productivity
- Maintenance support from Falcon-trained partner technicians
- Proactive planning to minimise downtime with preventative maintenance and anticipation of future needs through ongoing operational assessments
- Optimise system uptime, continuity of throughput and production efficiency

• Warranty

The system will be covered under standard **1-year comprehensive warranty**.

• Supply and management of spare parts.

On completion of the DAP, when the configuration will be finalized a final and detailed list, which will be extended to include all spare and wear parts, will be presented to UHP. To maintain the required uptime spares will be maintained at site.

• Spares Depot

Falcon has spares depot in ANZ and on-site spares will be replenished from this common pool.

• On Site Support

To adhere to the required System uptime, dedicated on- site team should be available at site covering all shifts. Falcon proposes that on site team to be on UHP's Payroll, trained by Falcon.

- **Hotline Support**

The hotline service access is available 24 X 7. The main function is to dispatch the call to the proper qualified engineer for immediate answer. If the Falcon qualified engineer is not available, the call is transferred to a back-up qualified engineer and finally an answering machine. In the last case, the UHP's personnel shall be contacted within the time frame described in the service level. The Falcon qualified engineer will support UHP maintenance personnel by using a remote access (via VPN) to the equipment/systems.

19. System Maintenance

19.1 Daily Check-

Detailed maintenance scheduled will be shared with UHP and daily checks to be performed by the site technicians.

19.2 Preventive Maintenance

Regular preventative maintenance as per schedule will be conducted systematically throughout the system. Site technicians will manage all operational tasks and preventive maintenance for covered items, with additional assistance from Falcon trained local partner technicians.

19.3 Corrective Maintenance

Corrective maintenance will be performed in conjunction with preventive maintenance activities. Any required corrective actions will be communicated to UHP for approval prior to execution, and scheduling will be coordinated accordingly. Charges will apply for any parts needed for corrective maintenance that are not covered by warranty.

20. Commercial Terms

20.1 Price Sheet for 4224 Tote locations

Price Schedule		
Pos.	Description	Amount (AUD)
FM1	Supply for NEO, Conveyors and Sorter Automation System	\$ 2,639,656.73
FS1	Project Management	\$ 87,525.08
FS2	Installation and Commissioning	\$ 965,728.72
FF1	Packing Charges	\$ 107,072.44
TOTAL		\$ 3,799,982.98

20.2 Price Sheet for 5544 Tote locations

**This option is created extrapolating the 3 weeks average to 4 weeks average.*

Price Schedule		
Pos.	Description	Amount (AUD)
FM1	Supply for NEO, Conveyors and Sorter Automation System	\$ 2,804,873.48
FS1	Project Management	\$ 95,217.88
FS2	Installation and Commissioning	\$ 1,114,226.45
FF1	Packing Charges	\$ 113,566.34
TOTAL		\$ 4,127,884.15

20.3 Price Sheet for Additional Conveyors

Price Schedule		
Pos.	Description	Amount (AUD)
FA1	Accumulation Conveyor for Bulk Material	\$ 70,256.14
FA2	Slide Gate Roller Conveyor	\$ 33,622.28
TOTAL		\$ 103,878.42

The Total Price is to be understood.

- Inco Terms- Ex-Works, Greater Noida, India.
- Installation & Commissioning- Included
- Freight- In UHP's Scope (from Falcon Factory to site)
- Taxes & duties as actual- In UHP's Scope
- Price is valid for 60 days from the date of proposal.

20.4 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 6.88% advance on contract signing.

Stage 2: 6.88% advance on completion of drawing approval process.

Stage 3: 34.41% along with 100% Taxes against PI prior to dispatch on Pro-rata basis.

Stage 4: 10.62% at commencement of installation.

Stage 5: 13.11% at installation midpoint.

Stage 6: 13.11% at completion of installation and commissioning

Stage 7: 15% retention for 30 days after Go Live.

**All invoices (if due as per payment terms) should be cleared within 7 days from the date of receipt of undisputed invoice by the buyer.*

21. Warranty Period

Falcon offers Comprehensive warranty for 1 year (Starts from the date of beneficiary use).

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)
- 6 Preventive Maintenance Visits per year and 6 Breakdown visits shall be covered under warranty. (Any other visits shall be charged Extra)

The warranty does not apply to the replacement or repair of:

- Normal wear and tears.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that is supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

22. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Construction Power**
- **Steel Works- If not specified**
- **Collection Bins**
- **Fans anywhere around the system.**
- **Lightning**
- **Carton Erector or any other packing equipment is not specified in the proposal.**
- **Hand-held scanners other than those mentioned in the proposal.**
- **Waste Disposal**
- **Laydown Area Fencing**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices.
- Traffic and route markings
- Unloading and Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.